



Czarny Kot Bookstore

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Introduction

The proposed system is an online bookstore that sells physical books in a wide range of genres and supports multiple physical formats, including hardcover, paperback, and boxed or special editions.

The software is created to replace fragmented manual processes such as spreadsheet-based stock tracking or manual e-mail order handling. A dedicated system reduces mistakes, shortens order handling time, and makes the customer experience more predictable.

Another motivation is commercial growth. A structured online channel increases the store's reach, supports promotions, and allows easier measurement of sales.



Main features

Customers can search the catalog, compare editions, place orders, pay online, and track delivery status from a single web application.

The system is designed for customers, store staff, and administrators. Customers need a fast and intuitive shopping experience, while the store needs reliable inventory control, order processing, shipping coordination, and reporting.



Technology used for the implementation

The project was implemented in the PrestaShop™ environment. A Docker-based architecture was used to eliminate configuration issues related to PHP extensions and communication with external APIs (DHL, PayU). This solution guarantees complete service isolation and environmental compatibility.



Software requirements specification

Functional Requirements

- **Product Management:** The administrator can add, edit, and delete products from the database. Each product must have a description, price, category, and SEO keyword support.
- **Purchase Process:** Users can add products to the cart, apply discount codes (with value validation), and finalize orders.
- **Payment and Logistics Integration:** The system has built-in interfaces for communicating with APIs from external payment service providers (PayU) and logistics providers (DHL) to process transactions and generate shipping labels.
- **User Accounts:** Users can register and login into a profile with saved address data, allowing for automatic form completion.



Software requirements specification

Non-Functional Requirements

- Responsiveness: The store interface is fully scalable and mobile-friendly (RWD).
- Environment: The application is containerized using Docker, ensuring a consistent environment between the developer and the production server.
- Performance: The system ensures correct handling of communication with external APIs in a sandbox environment without delays preventing the completion of the purchasing process.



Software requirements specification

Actors:

- Visitor - anonymous user who can browse the catalog and view book details.
- Customer - registered or logged-in user who can order books, pay and track shipments.
- Admin - staff member who manages catalog data, inventory, and orders.
- Payment Gateway - external service used to authorize and capture payments.
- Courier Service - external shipping provider used to create shipments and track delivery.

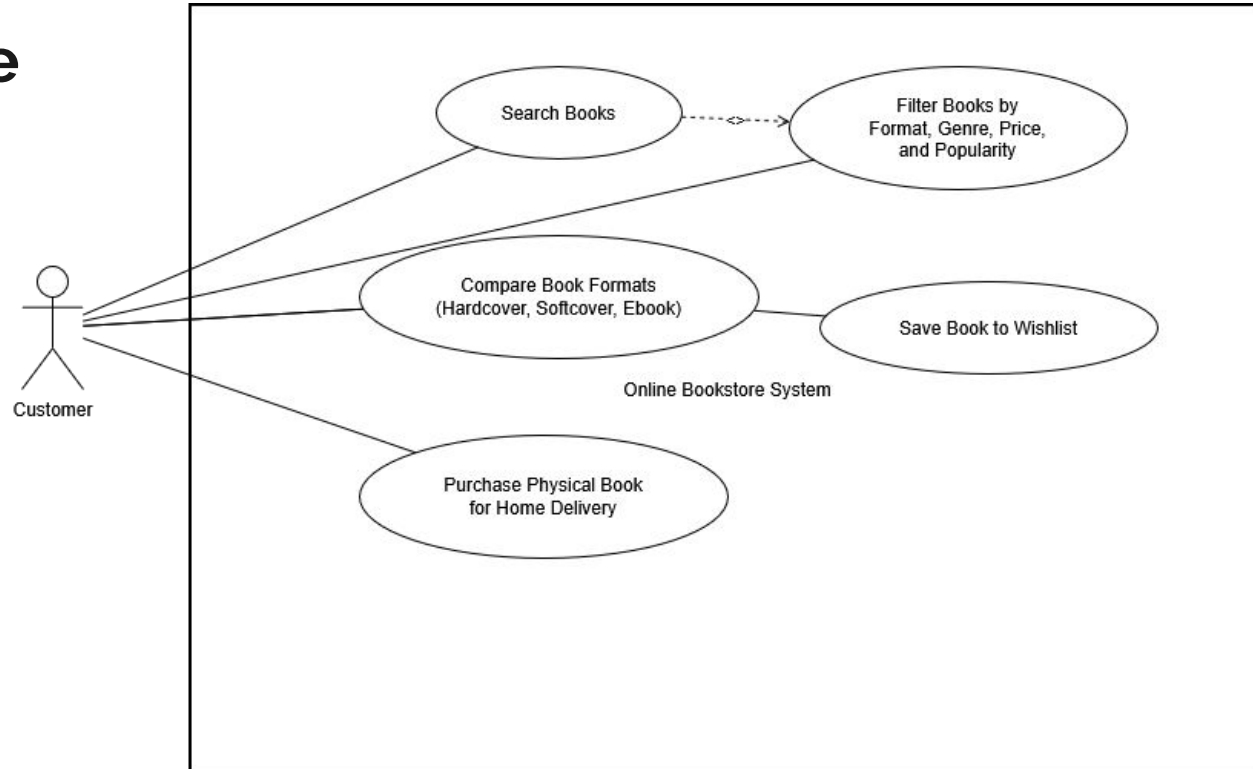


User story

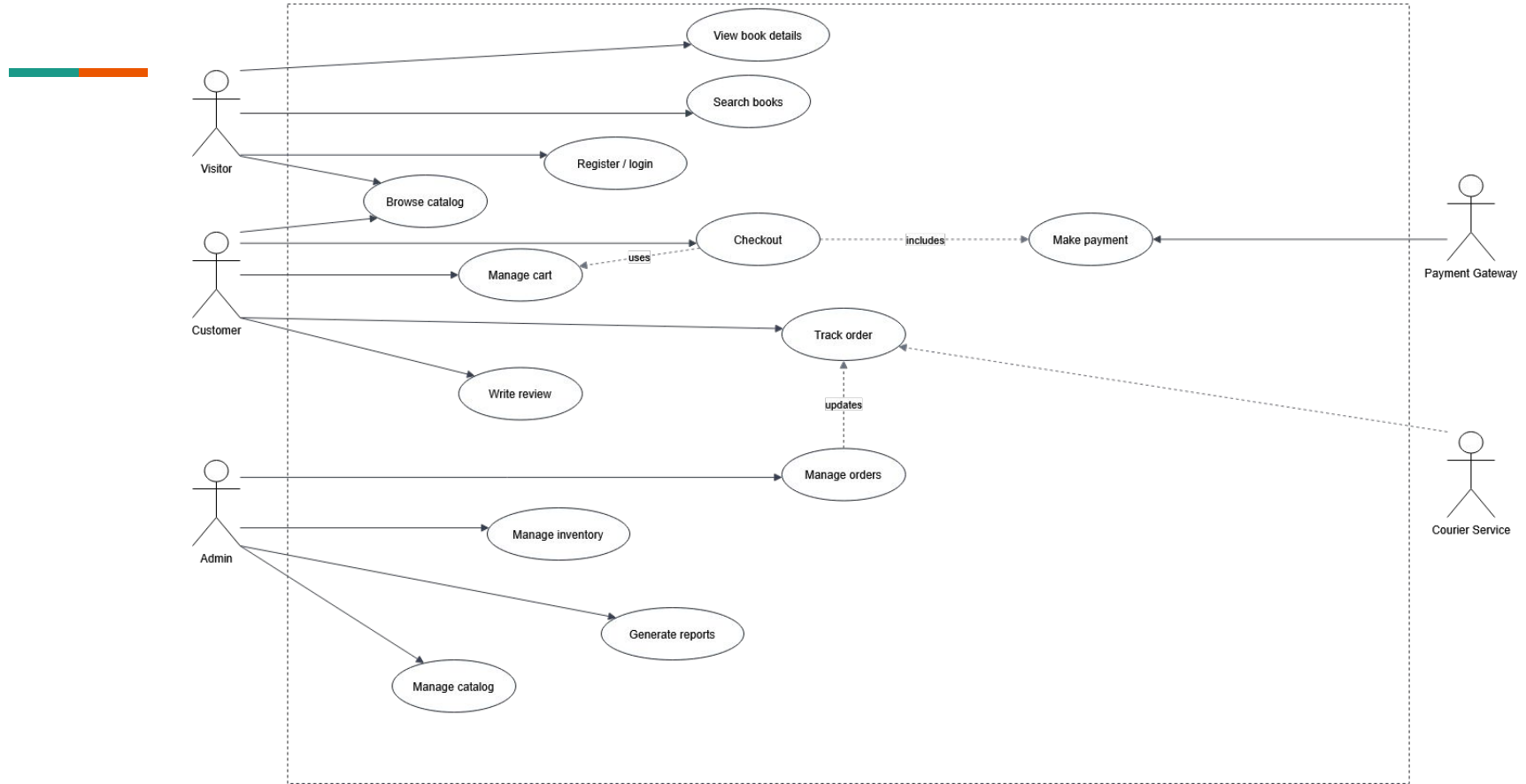
As a 25-year-old woman living in a mid-sized city with a range of various interests, I want to browse and buy softcover, hardcover versions of books in one place so that I can easily find titles that fit my mood, budget, and reading habits.

- I can search and filter books by format, genre, price, and popularity.
- I can compare hardcover, softcover options for the same title.
- I can buy physical books for home delivery.
- I can save books to a wishlist for later consideration.

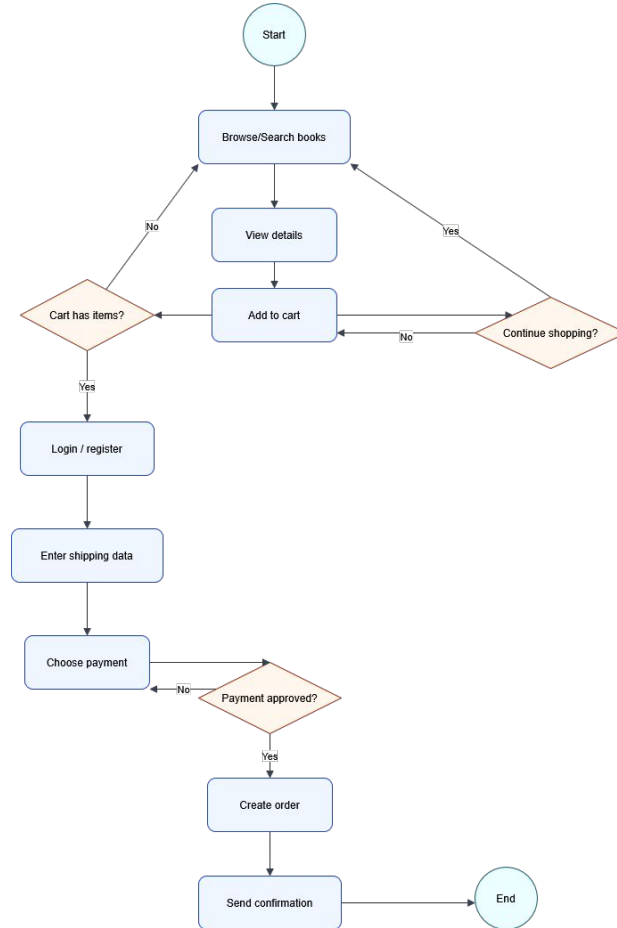
Typical use case



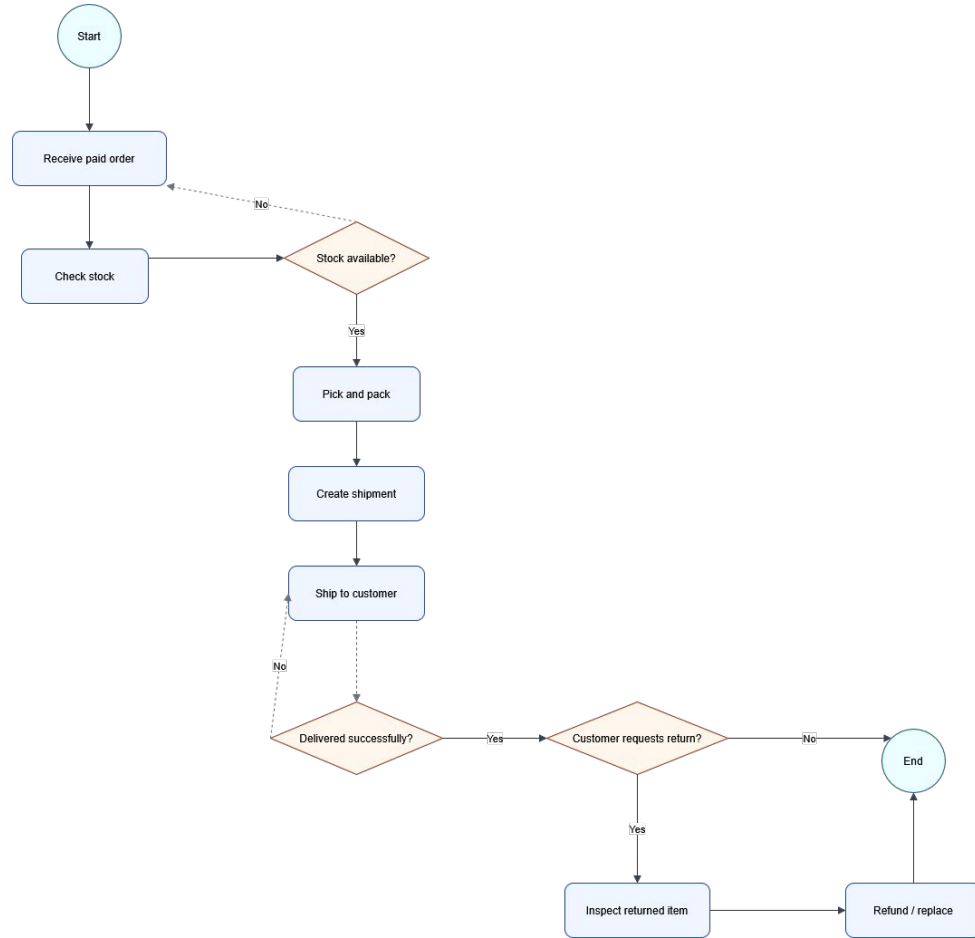
Combined use case diagram



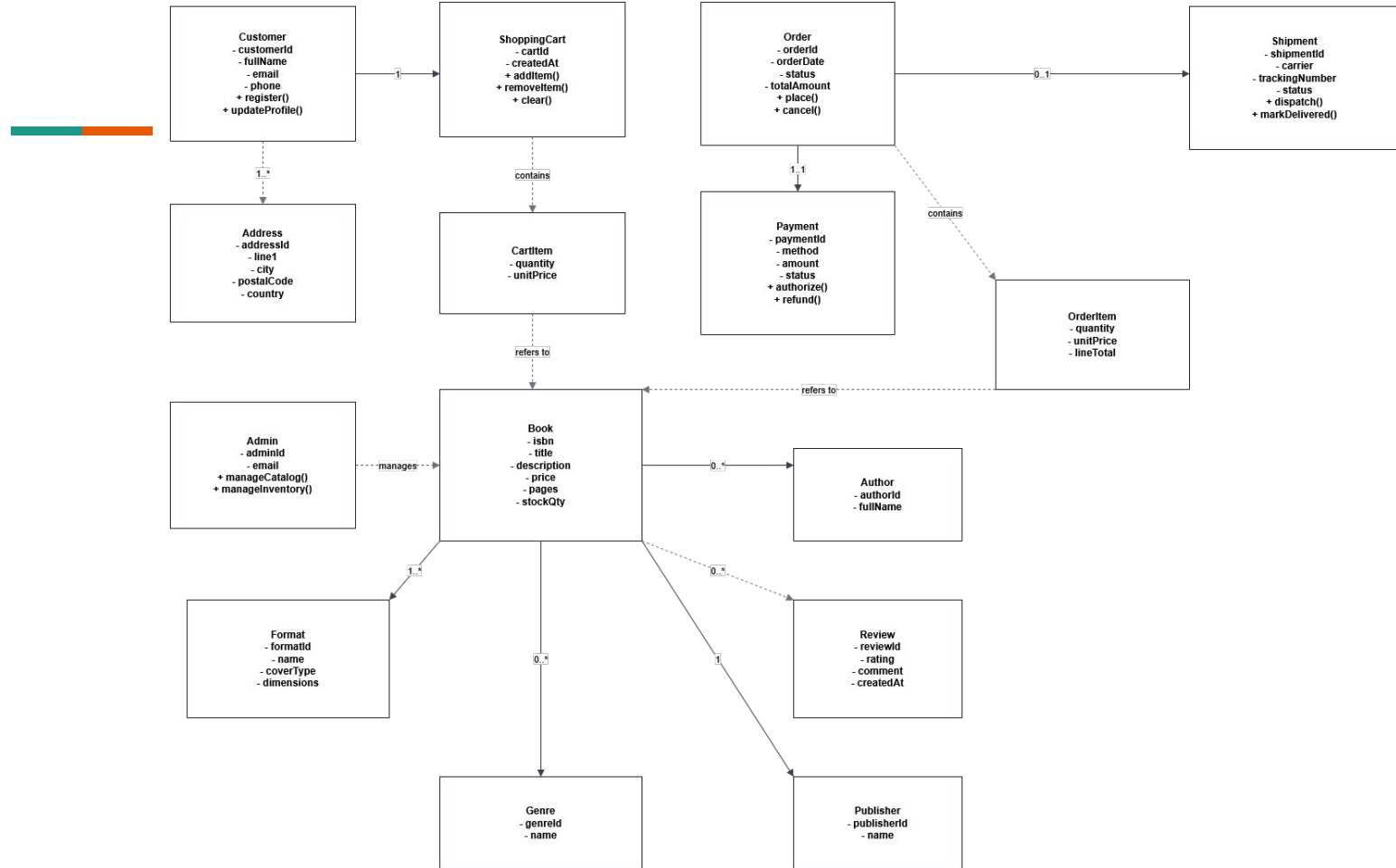
Activity diagram - Checkout



Activity diagram - Order fulfillment



Class diagram



Sequence diagram

